

Data Converters

Exercise 03 – Noise, Jitter, Track & Hold Circuit

Task I – The very “cool” resistor

The total integrated noise of any RC circuit results in kT/C . Shown in Figure 1 is a circuit and its small-signal model for $C_0/C_F = N \gg 1$ and $\omega \ll g_m/C_0$ that generates less voltage noise than $4kTR$. Consequently, the total integrated noise is less than kT/C if the circuit is combined with a capacitor. Let's analyze this very “cool” resistor.

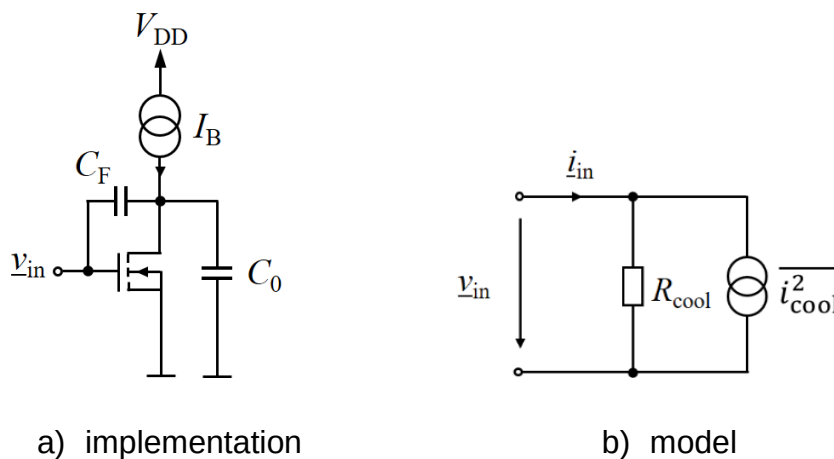


Figure 1: A very "cool" circuit (a) and its model for $C_0/C_F = N \gg 1$ and $\omega \ll g_m/C_0$ (b).

- Draw the small-signal equivalent circuit of Figure 1.a neglecting r_{ds} .
- Show that $R_{cool} \approx \frac{N}{g_m}$ for $C_0/C_F = N \gg 1$ and $\omega \ll g_m/C_0$.
- Draw the small-signal equivalent circuit including thermal noise and neglecting r_{ds} .
- Show that $\overline{i_{cool}^2} \approx \frac{1}{N} \frac{4kT}{R_{cool}}$.

- e. The circuit is now connected to a noisy signal source and a capacitor as shown in Figure 2 where $R_s = N \cdot R_{\text{cool}}$ and $C_0 \ll N \cdot C$.

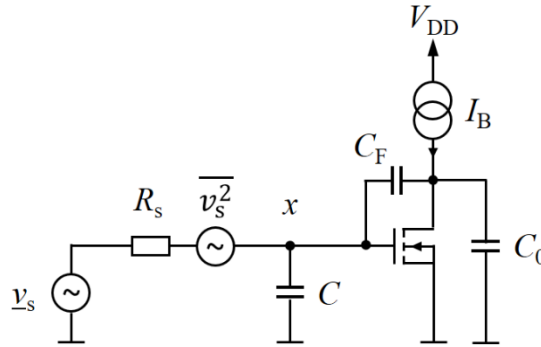


Figure 2: Combination of the very "cool" resistor with a noisy signal source and a capacitor.

Draw the small-signal equivalent noise model using the results of tasks T1.a to T1.d.

Note: Replace the voltage noise source $\overline{v_s^2}$ by its equivalent current noise source $\overline{i_s^2}$.

- f. Show that the total integrated noise voltage at node x results in

$$\overline{v_{x,\text{total}}^2} = \frac{2}{N} \cdot \frac{kT}{C}$$

Note: $R_s = N \cdot R_{\text{cool}}$ and $C_0 \ll N \cdot C$.

- g. Calculate g_m for $C = 1$ pF, $R = 1$ k Ω , and $N = 20$. Do you consider the circuit useful?

Note: $R_s = N \cdot R_{\text{cool}}$ and $C_0 \ll N \cdot C$.

Task II – SNR Degradation due to Jitter

You measure the SNR of a 10-bit ADC using a full-scale sinusoidal input signal. For low input frequencies, the result is $SNR \approx 62$ dB, as expected. However, applying an input frequency of 36 MHz, the SNR drops to 58 dB.

Assuming the cause to be clock jitter, what is the standard deviation σ_t of the clock jitter?

Task III – Error Analysis of a Track & Hold Circuit

A track & hold circuit and its timing diagram are shown in Figure 3.

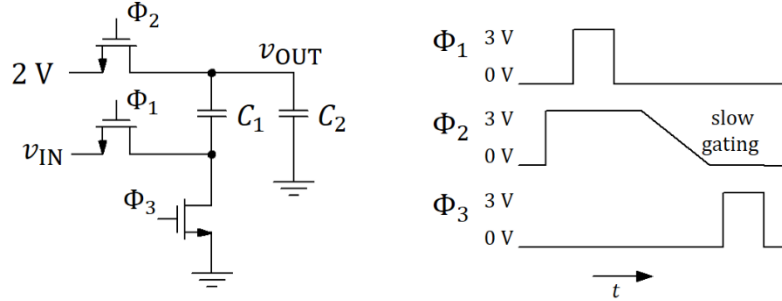


Figure 3: Track & hold circuit and its timing diagram.

Given are:

$$C_1 = 200 \text{ fF} \quad C_2 = 200 \text{ fF} \quad C_{\text{ox}} = 20 \text{ fF} \quad C_{\text{ovl}} = 2 \text{ fF} \quad V_{\text{th}} = 0.5 \text{ V}$$

- Apply the principle of charge conservation and determine $v_{\text{OUT}}(v_{\text{IN}})$ in phase $\Phi_3 = 3 \text{ V}$.
- For $v_{\text{IN}} = 1 \text{ V}$, the ideal output voltage results in $v_{\text{OUT}} = 1.5 \text{ V}$. Determine the deviation Δv_{OUT} from the ideal output voltage v_{OUT} due to charge injection and clock feedthrough when Φ_3 goes high.

notes:

All clock transients correspond to the case of “fast gating” except for Φ_2 going low, which corresponds to “slow gating”.

For simplicity, it can be assumed that the charge splits equally between source and drain in the case of “fast gating”.

- Calculate the kT/C -noise in μV_{rms} at the node v_{OUT} when Φ_3 goes low.

note:

Only the capacitors C_1 and C_2 are to be considered.